Ex.No.1A	AUTOMATICALLY GENERATE FUNCTION DEFINITIONS USING OpenAI GPT-3 API	URK23CS8018
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Aim:

To develop a Python application that generates and tests functions from natural language descriptions using GPT.

Procedure:

Step 1: Import the required OpenAI library.
Step 2: Initialize the OpenAI client using an API key.
Step 3: Get a natural language description of the required function from the user.
Step 4: Send the description to the GPT API with a prompt requesting valid Python code.
Step 5: Receive and display the generated Python function code.
Step 6: If the API is unavailable, generate predefined functions using an offline function generator.
Step 7: Execute the generated Python code using the exec() function.
Step 8: Test the generated function with predefined sample inputs.
Step 9: Compare the actual output with the expected output.
Step 10: Display whether the test has passed or failed.

Program:

```
from openai import OpenAI
client = OpenAI()
description = input("Describe the function: ")
try:
   prompt = f"""
Generate only valid Python code.

Generate exactly one Python function based on this description:

{description}

Do not use markdown.
Do not use ```python.
Only return executable Python code.
"""

   response =
      client.responses.create( model=
      "gpt-5.6-luna", input=prompt
   )

   code = response.output_text

   print("\n===== GPT Generated Code =====\n")
   print(code)
```

```
except Exception as e:
    print("\nOpenAI API is unavailable.")
    print("Reason:", e)
    print("\nSwitching to Offline Function Generator...\n")
    d = description.lower()
    if "add" in d:
        code = """
def add(a, b):
    return a + b
"""
    elif "subtract" in d:
        code = """
def subtract(a, b):
    return a - b
"""
    elif "multiply" in d:
        code = """
def multiply(a, b):
    return a * b
"""
    elif "divide" in d:
        code = """
def divide(a, b):
    return a / b
"""
    elif "square" in d:

        code = """
def square(x):
    return x * x
"""
    elif "cube" in d:
        code = """
def cube(x):
    return x * x * x
"""
    elif "factorial" in d:
        code = """
def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n - 1)
"""
    else:
        code = """
def hello():
```

```
return "Function not available"
"""
    print(code)
exec(code)
print("\n===== Testing Function =====")
d = description.lower()
if "add" in d:
    result = add(10, 20)
    expected = 30
elif "subtract" in d:
    result = subtract(20, 5)
    expected = 15
elif "multiply" in d:
    result = multiply(4, 5)
    expected = 20
elif "divide" in d:
    result = divide(20, 4)
    expected = 5
elif "square" in d:
    result = square(8)
    expected = 64
elif "cube" in d:
    result = cube(3)
    expected = 27
elif "factorial" in d:
    result = factorial(5)
    expected = 120
else:
    result = hello()
    expected = "Function not available"
print("Output:", result)
print("Expected:", expected)
if result == expected: print("\n
    Test Passed")
else:
    print("\n+  Test  Failed")
```

Output Screenshot:

```
PS C:\Users\LENOVO\downloads\AI> python main.py
Describe the function: Create a function to add two numbers

===== AI Generated Code =====

def add(a, b):
    """
    Adds two numbers and returns the result.

    Args:
        a (int): The first number to be added.
        b (int): The second number to be added.

    Returns:
        int: The sum of `a` and `b`.
    """
    return a + b

# Example usage
result = add(3, 5)
print(result)  # Output: 8
8

===== Testing Function =====

Output: 30
Expected: 30

Test Passed
```

```
PS C:\Users\LENOVO\downloads\AI> python main.py
Describe the function: Create a function to multiply two numbers

===== AI Generated Code =====

def multiply(a, b):
    """
    Multiply two integers and return the result.

    Args:
        a (int): The first integer.
        b (int): The second integer.

    Returns:
        int: The product of a and b.
    """
    # Check if both inputs are integers
    if not isinstance(a, int) or not isinstance(b, int):
        raise ValueError("Both arguments must be integers.")

    # Multiply the two numbers
    result = a * b

    return result

===== Testing Function =====

Output: 20
Expected: 20

Test Passed
```

```
PS C:\Users\LENOVO\downloads\AI> python main.py
Describe the function: Create a function to subtract two numbers

===== AI Generated Code =====

def subtract(x, y):
    return x - y

===== Testing Function =====

Output: 15
Expected: 15

Test Passed
```

```
PS C:\Users\LENOVO\downloads\AI> python main.py
Describe the function: Create a function to divide two numbers

===== AI Generated Code =====

def divide(a, b):
    if b == 0:
        raise ValueError("Cannot divide by zero")
    return float(a) / b

===== Testing Function =====

Output: 5.0
Expected: 5

Test Passed
```

```
PS C:\Users\LENOVO\downloads\AI> python main.py
Describe the function: Create a function to calculate the square of a number

===== AI Generated Code =====

3
def square(number):
    """
    Calculate the square of a given number.

    Args:
    number (int): The number whose square we want to calculate.

    Returns:
    int: The square of the input number.
    """
    return number ** 2

# Example usage:
result = square(4)
print(result)  # Output: 16
16

===== Testing Function =====

Output: 64
Expected: 64

Test Passed
```

```
PS C:\Users\LENOVO\downloads\AI> python main.py
Describe the function: Create a function to calculate the cube of a number

===== AI Generated Code =====

3
def cube(x):
    """
    Calculate the cube of a given number

    Args:
        x (float): The number whose cube is to be calculated

    Returns:
        float: The cube of the input number
    """
    return x ** 3

# Example usage:
print(cube(2))   # Output: 8.0
print(cube(5.5)) # Output: 167.6491450105171
8
166.375

===== Testing Function =====

Output: 27
Expected: 27

Test Passed
```

```
PS C:\Users\LENOVO\downloads\AI> python main.py
Describe the function: Create a function to calculate the factorial of a number

===== AI Generated Code =====

def factorial(n):
    if n == 0:
        return 1
    else:
        result = 1
        for i in range(1, n + 1):
            result *= i
        return result

# Example usage:
number = int(input("Enter a number: "))
result = factorial(number)
print(f"The factorial of {number} is: {result}")
Enter a number: 7
The factorial of 7 is: 5040

===== Testing Function =====

Output: 120
Expected: 120

Test Passed
```

Link:

https://drive.google.com/file/d/1TaMsdDn5UXK6ICB1AQYye35u13VUiR0w/view?usp=sharing

Result:

Thus, the application successfully generated, executed, and tested Python functions.